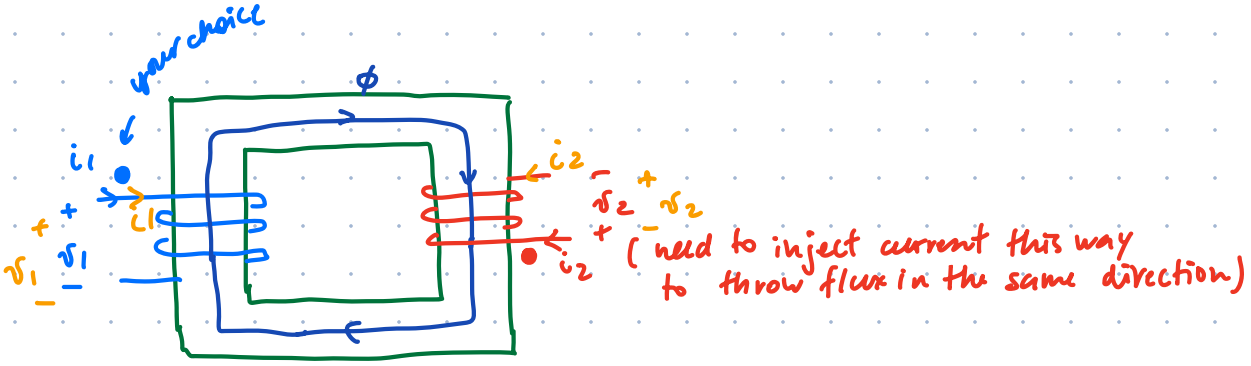
Review 4 (HW 4) 09/28/2026

Coupled coils

Dot convention: „ currents entering dots throw flux in the same direction".

We want our reference model to work irrespective of winding direction

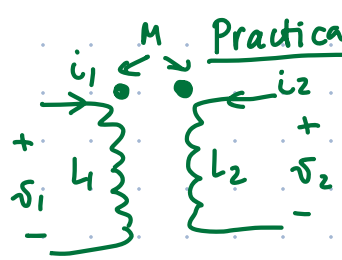


Ideal



$$\frac{i_1}{-i_2} = \frac{N_2}{N_1} \qquad \frac{v_1}{v_2} = \frac{N_1}{N_2}$$

Practical



$$\lambda_1 = N_1\phi_1 = L_1 i_1 + M i_2$$

$$\lambda_2 = N_2\phi_2 = M i_1 + L_2 i_2$$

$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

⇓ phasors:

$$\bar{V}_1 = j\omega \bar{I}_1 L_1 + j\omega \bar{I}_2 M$$

$$\bar{V}_2 = j\omega \bar{I}_1 M + j\omega \bar{I}_2 L_2$$

① Fix ref. frame
② Apply ref. model
③ Subst. back

Coefficient of coupling: $0 \le K \le 1$

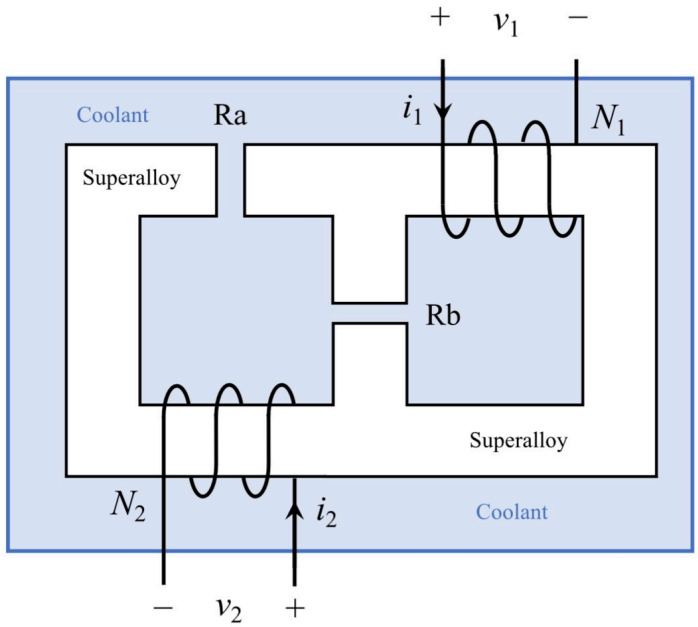
$$K = \frac{M}{\sqrt{L_1 L_2}}$$

① Spring 2021, M1, pr 4

Problem 4 (25 Points)

You have been hired by a stealth-mode startup to develop a superconducting transformer for a possible mission to Mars.

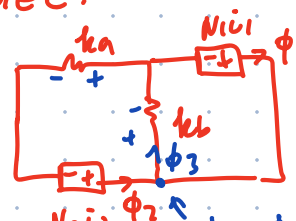
The transformer is built out of magnetic super alloy, and immersed in a liquid coolant. The super alloy is so magnetically permeable that its permeability is essentially infinite ($\mu_r \to \infty$) while the coolant is a factor of five times more permeable than vacuum ($\mu_r = 5$). The two coils are connected to a resonant power converter circuit with the voltage and current labels as shown. The first coil has $N_1 = 20$ turns while the second coil has $N_2 = 10$ turns.



a) We computed Ra = 200 At/Wb and Rb = 800 At/Wb by neglecting the effects of fringing. Use these figures to compute the mutual inductance between the two coils, and state the unit.

M = 1H

① MEC:



KVL: $-\phi_2 k_a - N_1 i_1 + N_2 i_2 = 0$

$\phi_3 = \phi_2 + \phi_1$

$\phi_1 = \phi_3 - \phi_2$

$$\phi_2 = \frac{N_2 i_2 - N_1 i_1}{k_a}$$

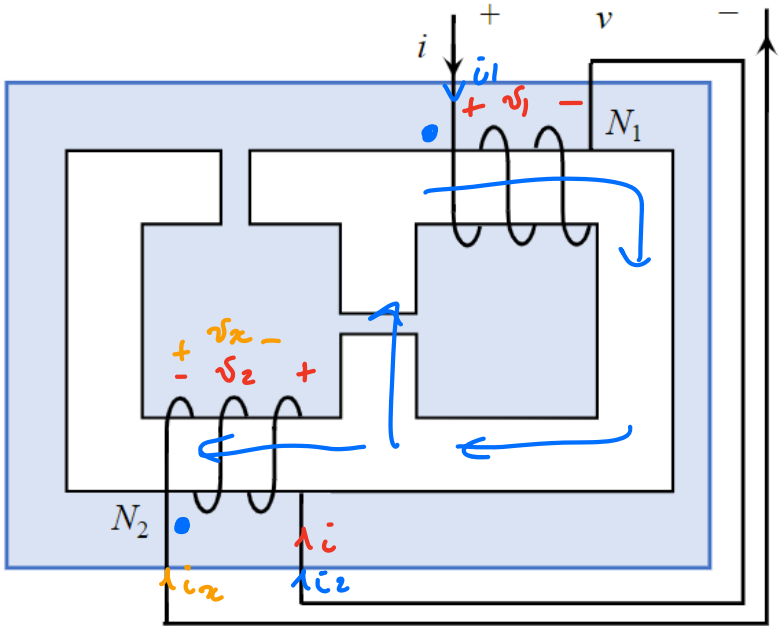
$$\lambda_1 = N_1\phi_1 = \left(\frac{N_1^2}{k_b} + \frac{N_1^2}{k_a}\right) i_1 - \frac{N_1 N_2}{k_a} i_2$$

$L_1 = 2.5\,H$

$$\lambda_2 = N_2\phi_2 = \frac{N_2^2}{k_a} i_2 - \frac{N_1 N_2}{k_a} i_1$$

$L_2 = 0.5\,H$

(c)



In part c, two coils are connected in series as shown. What is the combined inductance of the structure?

$L_1 = 2.5\,H$

$L_2 = 0.5\,H$

$M = 1\,H$



- $v = L_{eq}\frac{di}{dt}$ (L_{eq} = ?)
- $v = v_1 + v_2$

$i = i_1 = i_2$

Ref: $v_1 = L_1\frac{di_1}{dt} + M\frac{di_x}{dt}$

$v_x = M\frac{di_1}{dt} + L_2\frac{di_x}{dt}$

$i_x = -i_2$

$v_x = -v_2$

⇓

Sub:

$v_1 = L_1\frac{di}{dt} - M\frac{di}{dt}$

$v_2 = -M\frac{di}{dt} + L_2\frac{di}{dt}$

$\Rightarrow v = \underbrace{(L_1 + L_2 - 2M)}_{L_{eq} = 1H}\frac{di}{dt}$